CODE:

```
:- dynamic fact/1.

% Initial facts

fact(a).

fact(b).

% Rules

rule(c,[a,b]).

rule(d,[c]).

rule(e,[d]).

% Check conditions

check_conditions([]).

check_conditions([H|T]) :-

  fact(H),

  check_conditions(T).

% Forward chaining

forward_chain :-

  rule(NewFact,Conditions),

  \+ fact(NewFact),

  check_conditions(Conditions),

  assertz(fact(NewFact)),

  write('Derived: '), write(NewFact), nl,

fail.

forward_chain.
```

OUTPUT:

```
53 ?- consult(['c:/Users/arunp/OneDrive/Documents/AI/EXP 29.pl']).
true.

54 ?-
|    consult(['c:/Users/arunp/OneDrive/Documents/AI/EXP 29.pl']).
true.

54 ?- forward_chain.
Derived: c
Derived: d
Derived: e
true.

55 ?- fact(c).
true.
```